

[K S Tharunpranav 251801359-2025-AIDS](#) ▾**K2****Started on** Monday, 24 August 2026, 9:01 PM**State** Finished**Completed on** Monday, 24 August 2026, 9:01 PM**Time taken** 25 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int find_first_zero(int *arr, int low, int high) {
5      if (high >= low) {
6          int mid = low + (high - low) / 2;
7
8          if ((mid == 0 || arr[mid - 1] == 1) && arr[mid] == 0) {
9              return mid;
10         }
11
12         if (arr[mid] == 1) {
13             return find_first_zero(arr, mid + 1, high);
14         } else {
15             return find_first_zero(arr, low, mid - 1);
16         }
17     }
18     return -1;
19 }
20
21 int main() {
22     int m;
23     if (scanf("%d", &m) != 1) return 0;
24
25     int *arr = (int *)malloc(m * sizeof(int));
26     if (arr == NULL) return 0;
27
28     for (int i = 0; i < m; i++) {
29         if (scanf("%d", &arr[i]) != 1) {
30             free(arr);
31             return 0;
32         }
33     }
34
35     int first_zero_index = find_first_zero(arr, 0, m - 1);
36
37     if (first_zero_index == -1) {
38         printf("0\n");
39     } else {
40         printf("%d\n", m - first_zero_index);
41     }
42
43     free(arr);
44     return 0;
45 }
46

```

	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓
✓	10 1 1 1 1 1 1 1 1 1 1 1	0	0	✓
✓	8 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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